

Session 1:

Java Basics – Introduction, History and Features of Java, Software Requirements, Java Installation

Session 2:

IDE – NetBeans IDE, Eclipse IDE Overview and Usage

Session 3:

Java Architecture – JVM Architecture, Java Program Structure, Basic Syntax

Session 4:

Variables – Syntax, Data Types, Tokens, Operators, Introduction to OOP Concepts

Session 5:

Branching and Looping – Control Structures, Selection Statements, Looping Controls

Session 6:

Functions and Arrays – User Defined Functions, Predefined Functions, Arrays

Session 7:

Classes and Objects – Classes and Objects, Static Variables, Static Methods, Static Blocks

Session 8:

Constructors – Constructor Types, Constructor Overloading

Session 9:

Features of OOPs – Inheritance, Polymorphism, Method Overriding, Abstraction

Session 10:

Features of OOPs – Super Keyword, This Keyword, Encapsulation, Data Binding, Message Passing

Session 11:

Packages – java.lang, java.util, java.math, java.io, java.net Packages

Session 12:

String and Its Functions – String Class, StringBuilder, StringBuffer

Session 13:

Exception Handling – Predefined Exceptions, User Defined Exception Handling

Session 14:

Collection Framework – ArrayList, LinkedList, Vector, Stack

Session 15:

Input and Output – FileInputStream, FileOutputStream

Session 16:

Excel Handling – Reading Data from Excel, Writing Data into Excel

Session 17:

Lab – Java Basic Programs Practice

Session 18:

Automation Testing – Introduction to Automation, Introduction to Selenium, WebDriver Interface Methods

Session 19:

Selenium Locators – Locators, Select Class, Actions Class

Session 20:

Advanced Selenium – JavaScript Executor, Waits in Selenium

Session 21:

Selenium Alerts – Handling Alerts in Selenium

Session 22:

Advanced Selenium Tools – Java Robot Class, Selenium Grid Architecture, Hub and Node Configuration

Session 23:

Frameworks – Data Driven Testing, Database Testing

Session 24:

Frameworks – Keyword Driven Testing, Hybrid Testing

Session 25:

TestNG Framework – Introduction, Annotations, Execution Flow

Session 26:

TestNG Framework – Continued, Reports, Assertions, Parallel Execution

Session 27:

Cucumber – Cucumber Framework, Maven Integration

Session 28:

API Testing using Postman – Introduction to Web Services, Postman Installation, Components

Session 29:

Postman Operations – Navigation, Sending Requests, Creating Collections

Session 30:

HTTP Methods – GET, POST, PUT, DELETE, Response Handling in Postman

Session 31:

Mobile Testing – Introduction to Mobile Testing, Appium Architecture, Features of Appium, How Appium Works